

Flatcar Nebraska

Nebraska is a self-hosted update management server for Flatcar Container Linux. It implements the **Omaha protocol** (same as ChromeOS updates) and gives operators control over which nodes update, when, and at what rate.

What Nebraska Provides

- Dashboard to monitor Flatcar instances and their update status
- Instance groups with configurable update channels (alpha/beta/stable or custom)
- Control over update rollout rate, time windows, and maximum simultaneous updates
- Machine alias display for easier identification
- Support for OEM extension payloads (required for OEM sysexec images since Flatcar 3760.0.0)

Pointing Flatcar Instances at Nebraska

New machines — via Butane:

```
variant: flatcar
version: 1.0.0
storage:
  files:
    - path: /etc/flatcar/update.conf
      overwrite: true
      mode: 0644
      contents:
        inline: |
          GROUP=stable
          SERVER=http://your.nebraska.host:8000/v1/update/
```

Custom group (defined in Nebraska dashboard):

```
contents:
  inline: |
    GROUP=ab51a000-02dc-4fc7-a6b0-c42881c89856
    SERVER=http://your.nebraska.host:8000/v1/update/
```

Existing machines — edit `/etc/flatcar/update.conf` :

```
echo "SERVER=http://your.nebraska.host:8000/v1/update/" >>
/etc/flatcar/update.conf
update_engine_client -update # force immediate check
```

Note: Default Nebraska sample containers use port 8000 with plain HTTP. Configure TLS for production.

Machine Alias

An optional per-instance name shown in the Nebraska dashboard instead of the instance ID:

```
storage:
  files:
    - path: /etc/flatcar/update.conf
      contents:
        inline: |
          SERVER=http://your.nebraska.host:8000/v1/update/
          MACHINE_ALIAS=prod-worker-01
```

Authorization

Nebraska supports OAuth2/OIDC for dashboard access. See `nebraska/authorization` for configuration details.

Development and Self-Hosting

Nebraska is open source: <https://github.com/flatcar/nebraska>

- Runs as a single binary or container
- Requires a PostgreSQL database
- The Nebraska dashboard exposes a web UI and REST API
- Supports i18n for non-English deployments

Important: Operators running their own Nebraska server must use a version that supports OEM sysext payloads (introduced in Flatcar 3760.0.0) to properly serve OEM extension updates.

OEM Payloads

Since Flatcar 3760.0.0, OEM vendor tools are distributed as sysext images (not baked into the OEM partition directly). Nebraska must serve these OEM payloads alongside the core OS

update. This means self-hosted Nebraska deployments need to be kept up to date to serve the correct OEM sysexr for each Flatcar release.